GeneWars v0.9 — Game Manual

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0. Basic Information

Item	Details
Name	GeneWars
Engine	Rebuilt on an open-source tower-defense engine
Genre	Tower defense + synthetic-biology educational metaphor
Run	Windows; unzip and keep the whole folder together when running
Status	v0.9 tester build

1. Background & Objectives

This game turns core concepts of synthetic biology (ATP as the energy currency, metabolic burden, genetic circuits, CRISPR gene editing, RNA interference, antimicrobial resistance, biofilms, horizontal gene transfer, quorum sensing, and more) into mechanics the player can operate by hand. It is made for iGEM's **HP (stakeholder engagement)** and **Education (science communication)** tracks.

- The infected side = environmental pressure that was enhanced after a biochemical experiment went out of control;
- The plant side = a scientist using ATP agents to engineer and enhance the plants through gene editing.

The goal is not to "be like" any classic tower-defense game, but to let players intuitively grasp the logic of synthetic biology while playing.

2. Running & Distribution

This section explains how to launch the game, how saves are managed, and the asset principles used in this version.

- **Launch**: unzip, keep the folder intact, then double-click the launch script. The main program and resource pack must stay in the same directory.
- **Saves**: progress is stored in a local user-data directory; double-click the reset script to clear progress and start over.
- **Portable distribution**: in portable mode the config is written inside the local folder, which suits tester distribution.
- **Asset principles**: all art, sound and UI in this version are original to the team or CC0, and can be shown publicly.

3. Story Background

The infected population began with a biochemical experiment that spun out of control: they underwent a collective biological transformation and gene editing. Their broken limbs became the very medium of infection and enhancement.

Pathogens seeping from severed arms turn infected into fast, green slime forms. Biofilms grow on helmets and bucket armour, and even a discarded piece of gear can be picked up by a passing infected, giving rise to a new enhanced individual. They are now advancing on the lawn outside the lab.

A scientist has extracted ATP agents from plant metabolic pathways, hoping to arm the plants with the same gene-editing methods. Green ATP boosts a plant's expression efficiency; red ATP grants extreme enhancement, but at the cost of cellular collapse — the plant dies after one burst of over-expression. For the ice, doom and hypno area fungi, the scientist has also granted gene-editing abilities, letting them rewrite the traits of plants or infected within range in a single shot.

Use ATP in moderation. Between enhancement and death lies only a single overdose of expression.

4. Core Mechanics

This section breaks down the three core mechanics — ATP, fungus area gene-editing, and infected variants — and notes their synthetic-biology meaning.

4.1 ATP system (core interaction: click, then drag onto a plant)

Type	Effect	Synthetic-biology meaning
Green ATP	Pea-shooting plant attack speed ×1.3 – 1.5; the sun-producing fungus plant grows instantly	Raised substrate/inducer → up-regulated metabolic flux, stronger expression (safe)
Red ATP	Pea-shooting plant fires a huge burst, then dies; the sun-producing fungus plant produces a lot of sun, then dies	Over-expression → metabolic overload, resource exhaustion → cellular collapse

4.2 Fungus area gene-editing

- Plants: **ice, doom and hypno area fungi**
- Effect: gene-edits plants within range (shots gain freeze / higher damage); turns every infected within range into a **normal infected** (reversion).
- Meaning: area-wide genetic modification / trait rewriting / reversion mutation (a simplified metaphor).

4.3 Infected variants

Infected	Modification
Normal infected	Infected and broken apart; the severed arm leaks pathogens → turns into a green slime form with higher speed
Helmeted infected	Helmet carries a biofilm casing; when it drops, it spawns a second normal infected on the spot (high-yield one-shot plants' fire damage does not spawn an extra one)
Bucket-armoured infected	Bucket armour has a spiked biofilm; when it drops, it rolls forward and damages the first plant in its path
Pole-vaulting infected	Pole takes on a green mold texture; after one use it returns to hand until death

Shield infected	Reskin only; mechanics unchanged
Heavy-armour infected	Leaves behind a suit of armour on death; a normal infected picks it up → becomes a red-green shield infected (horizontal gene transfer)
Newspaper-carrying / gene-edit infected	Instantly kills the first plant it touches; if left alive too long it weakens the whole column; when killed, nearby infected turn green and speed up
Commander infected	Reskinned as a commander; continuously summons heavy-armour / newspaper / pole-vaulting / bucket-armoured infected (quorum sensing)

5. Mechanics → Synthetic-Biology Concept Mapping

Game mechanic	**Real-world concept**
Green ATP	Higher substrate concentration / up-regulated metabolic flux / stronger expression
Red ATP	Over-expression / metabolic burden / plasmid loss / cellular collapse
Fungus area gene-editing	CRISPR / area-wide gene modification / off-target effects / reversion mutation
Infected severed-arm pathogens	Pathogen infection / phenotype change
Helmet biofilm casing	Biofilm as protective carrier / carrying viable cells
Bucket-armour spiked biofilm	Biofilm mechanical properties / pathogenicity
Pole-vaulting green mold	Fungi / mold materials / biodegradation and reuse
Armour being picked up	Horizontal gene transfer / phenotype acquisition

Commander summons	Quorum sensing / regulation of collective behaviour

6. Controls & Getting Started

1. **Collect sun**: sun-producing plants / fungus plants produce sun (green ATP speeds up the fungus plant).
2. **Pick and plant**: click a plant card → click a lawn tile to plant; the shovel removes a plant.
3. **Use ATP**: click the ATP card → **drag it onto a target plant** to trigger it. Green is safe; red is extreme but lethal.
4. **Area editing**: plant an ice / doom / hypno area fungus and trigger it to gene-edit targets within range.
5. **Handle the infected**: adjust your plant layout and ATP usage to the infected variants (biofilms, summons, reversion).

7. Suggestions for Educators

- **Target audience**: middle/high-school students, undergraduates without a life-science background.
- **Learning goals**: after playing, students can explain "the basic components of a genetic circuit," "the main stresses on a chassis cell," and "why metabolic burden is a real problem."
- **Pre/post test**: a short 3–5 question quiz (e.g. Why does red ATP kill the plant? What do biofilms mean in reality?), comparing the accuracy before and after play.
- **Use cases**: classroom demos, club activities, science-museum booths, iGEM in-person workshops.
- **Discussion questions**: Why does red ATP kill the plant? How are off-target effects of gene editing reflected in the game?

8. Asset & IP Statement

- The **synthetic-biology mechanics, narrative and educational design are original to the team**.
- All **art, sound and UI in this version are original or CC0 assets**, containing no third-party characters or game assets.

- Gameplay is inspired by classic tower-defense games; this is an **idea-level** acknowledgement only. Characters and assets have been de-branded, and no third-party character names or likenesses are used.
- The engine is an open-source reimplementation, licensed separately.

9. To-Dos

- Expand the library of original assets and sound effects.
- Internal testing has been completed; the public launch date will be confirmed later.
- Record gameplay/operation videos and organize in-game screenshots for promotional use.